PROGRAMMING IN JAVA

CSA0915

ASSIGNMENT-CO1

CAMPUS SMART-PARKING SYSTEM

TASK 04 – Classes, Objects and Constructors

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Task04.md

Aim

To design classes and objects for the Campus Smart-Parking System using Java constructors.

Classes Identified

1. **ParkingSlot** – Stores slot number, vehicle type and availability.
2. **Vehicle** – Stores vehicle number and vehicle type.
3. **Ticket** – Stores ticket ID, entry time and exit time.
4. **ParkingFacility** – Manages parking slot allocation and release.

Constructors

A constructor initializes an object when it is created. It has the same name as the class and does not have a return type.

Types of Constructors

- Default constructor
- Parameterized constructor

Object Creation

Objects are created using the new keyword.

Conclusion

Classes organize the parking system into separate components. Constructors initialize object data and make the system easier to manage.

ParkingSlot.java

```
public class ParkingSlot {
    private int slotNumber;
    private String vehicleType;
    private boolean available;

    // Default constructor
    public ParkingSlot() {
        slotNumber = 0;
        vehicleType = "Not Assigned";
        available = true;
    }

    // Parameterized constructor
    public ParkingSlot(int slotNumber, String vehicleType) {
        this.slotNumber = slotNumber;
        this.vehicleType = vehicleType;
        available = true;
    }

    public void displayDetails() {
        System.out.println("Slot Number: " + slotNumber);
        System.out.println("Vehicle Type: " + vehicleType);
        System.out.println("Available: " + available);
        System.out.println();
    }

    public static void main(String[] args) {
        ParkingSlot slot1 = new ParkingSlot();
        ParkingSlot slot2 = new ParkingSlot(101, "Car");
        ParkingSlot slot3 = new ParkingSlot(102, "Two-Wheeler");

        System.out.println("Default Constructor:");
        slot1.displayDetails();

        System.out.println("Parameterized Constructor:");
        slot2.displayDetails();
        slot3.displayDetails();
    }
}
```

Output:

